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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

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Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
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- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

1 Scope

The present document specifies and contains the derivation of Test Points for NR RF test cases, thereby 3GPP TSG RAN WG5 will have a way of storing the input contributions provided. The test cases are described in TS38.521-1[2], TS38.521-2[3] and TS38.521-3[4],

The test cases which have been analysed to determine Test Points are included as .zip files.

The present document is applicable from Release 15 up to the release indicated on the front page of the present Terminal conformance specifications.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 38.521-1: "NR; UE conformance specification; Radio transmission and reception; Part 1: NR range 1".

[3] 3GPP TS 38.521-2: "NR; UE conformance specification; Radio transmission and reception; Part 2: NR range 2".

[4] 3GPP TS 38.521-3: "NR; UE conformance specification; Radio transmission and reception; Part 3: NR interworking between NR range1 + NR range2 and between NR and LTE".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

Other definitions used in the present document are listed in 3GPP TS 38.521-1 [2], 3GPP TS 38.521-2 [3] or 3GPP TS 38.521-3 [4].

Editor’s note: intended to capture definitions

3.2 Symbols

Symbols used in the present document are listed in 3GPP TR 21.905 [1], 3GPP TS 38.521-1 [2], 3GPP TS 38.521-2 [3] or 3GPP TS 38.521-3 [4].

Editor’s note: intended to capture definitions

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

Other abbreviations used in the present document are listed in 3GPP TS 38.521-1 [2], or 3GPP, 3GPP TS 38.521-1 [2], 3GPP TS 38.521-2 [3] or 3GPP TS 38.521-3 [4].

Editor's note: intended to capture definitions

4 Test coverage analysis

This clause contains information on test point analysis and test point selection for RX and TX test configuration tables in [2], [3] and [4]. The test point analysis should include selection of:

- Test environment
- Test frequencies
- Test channel bandwidth
- Test Subcarrier Spacing (SCS)
- Downlink configuration including modulation and RB allocation
- Uplink configuration including modulation and RB allocation
- Number of test points

4.1 Test point analysis for FR1

This clause contains information on test point analysis and test point selection for test cases in [2] clause 6 and 7 with information about transmitting test point selection for FR1 listed in table 4.1-1 and receiver test point selection in table 4.1-2.

Table 4.1-1: NR UE transmitter test point selection for FR1

Subclause	Number of test points	Justification in attachment	Comments
6.2.1 UE maximum output power	TBD	"38.521-1_TPanalysis_6.2.1_MaxOP.zip"	RAN5#2-5G-NR Adhoc
6.2.2 Maximum Power Reduction (MPR)	power class 3: 1040 power class 2: 920	"38.521-1_TPanalysis_6.2.2_MPR.zip"	RAN5#81
6.2C.1 Configured UE transmitted Output Power	270	"38.521-1_TPanalysis_6.2C.1_ConfigOPSUL.zip"	RAN5#80
6.2.3 UE A-MPR	Table 4.1.1.1-1	Table 4.1.1.1-1	See Table 4.1.1.1-1
6.3.1 Minimum output power	TBD	"38.521-1_TPanalysis_6.3.1_MinOP.zip"	RAN5#2-5G-NR Adhoc
6.3.3.2 General ON/OFF time mask	TBD	"38.521-1_TPanalysis_6.3.3.2_OnOff_M.zip"	RAN5#2-5G-NR Adhoc
6.3.4.2 Absolute power tolerance	6	"38.521-1_TPanalysis_6.3.4.2_AbsPtol.zip"	RAN5#78
6.3.4.3 Relative power tolerance	TBD	"38.521-1_TPanalysis_6.3.4.3_RelPtol.zip"	RAN5#81
6.3.4.4 Aggregate power tolerance	PUCCH: 6 PUSCH: 6	"38.521-1_TPanalysis_6.3.4.4_AggPtol.zip"	RAN5#78
6.4.1 Frequency error	5	"38.521-1_TPanalysis_6.4.1_FreqErr_v2.zip"	RAN5#2-5G-NR Adhoc RAN5#80
6.4.2.1 Error Vector Magnitude	PUSCH: 252 PUCCH: 36 PRACH: 36	"38.521-1_TPanalysis_6.4.2.1_EVM.zip"	RAN5#80
6.4.2.2 Carrier leakage	3	"38.521-1_TPanalysis_6.4.2.2_CarrLeak.zip"	RAN5#80
6.4.2.3 In-band emissions	36	"38.521-1_TPanalysis_6.4.2.3_IE.zip"	RAN5#80
6.4.2.4 EVM equalizer spectrum flatness	90	"38.521-1 TPanalysis_6.4.2.4_EVMequalizerSpectrumFlatness_v2.zip"	RAN5#3-5G-NR
6.4.2.5 EVM equalizer spectrum flatness for Pi/2 BPSK	45	"38.521-1 TPanalysis_6.4.2.5_EVMequalizerSpectrumFlatness_BPSK.zip"	RAN5#81
6.5.1 Occupied bandwidth	10	"38.521-1_TPanalysis_6.5.1_OccBW.zip	RAN5#78
6.5.2.2 Spectrum Emission Mask	144 for Power Class 3 144 for Power Class 2	"38.521-1_TPanalysis_6.5.2.2_SEM_v2.zip"	RAN5#3-5G-NR-Adhoc
6.5.2.4.1 NR Adjacent channel leakage ratio	920 for Power Class 3 920 for Power Class 2	"38.521-1_TPanalysis_6.5.2.4_ACLR_v2.zip"	RAN5#3-5G-NR-Adhoc
6.5.2.4.2 UTRA Adjacent channel leakage ratio	680	"38.521-1_TPanalysis_6.5.2.4.2_UTRA ACLR.zip"	RAN5#3-5G-NR Adhoc
6.5.3.2 Spurious emissions for UE co-existence	TBD	"38.521-1_TPanalysis_6.5.3.2_SEcoex.zip"	RAN5#2-5G-NR Adhoc
6.5.4 Transmit intermodulation	8	"38.521-1_TPanalysis_6.5.4_TxIm.zip"	RAN5#80

Table 4.1-2: NR UE receiver test point selection for FR1

Subclause	Number of test points	Justification in attachment	Comments
7.3 Reference sensitivity power level	45	"38.521-1_TPanalysis_7.3_RefSense.zip"	RAN5#79
7.4 Maximum input level	6	"38.521-1_TPanalysis_7.4_Maximun input level.zip"	RAN5#81
7.5 Adjacent Channel Selectivity	3	"38.521-1_TPanalysis_7.5_ACS.zip"	RAN5#2-5G-NR Adhoc
7.6.2 In Band Blocking	3	"38.521-1_TPanalysis_7.6.2_InB_Block.zip"	RAN5#2-5G-NR Adhoc
7.6.3 Out-of-band blocking	3	"38.521-1_TPanalysis_7.6.3_OobBlocking.zip"	RAN5#80
7.8.2 Wide band Intermodulation	3	"38.521-1_TPanalysis_7.8.2_WidebandIntermod.zip"	RAN5#81
7.9 Spurious emissions	3	"38.521-1_TPanalysis_7.9_RxSpurious.zip"	RAN5#81

4.1.1 Test point selection for FR1 in A-MPR test cases

4.1.1.1 A-MPR test cases for single carrier

This section contains information on test point selection for test case 6.2.3 in [2], Additional Maximum Power Reduction (A-MPR).

Selection of test points should include some possible worst combinations based on the A-MPR characteristics specified for each NS value and these shall be selected so that they match with corresponding spectrum emission requirements test points. The number of test points should be realistic.

Table 4.1.1.1-1: NS value specific test points for A-MPR single carrier

NS label	Number of test points	Justification	Comments
NS_04	139	"38.521-1_TPanalysis_6.2.3_AMPR_NS_04.zip"	RAN5#81
NS_35	112	"38.521-1_TPanalysis_6.2.3_AMPR_NS_35.zip"	RAN5#80

4.2 Test point analysis for FR2

This clause contains information on test point analysis and test point selection for test cases in [3] clause 6 and 7 with information about transmitting test point selection for FR2 listed in table 4.2-1 and receiver test point selection in table 4.2-2.

Table 4.2-1: NR UE transmitter test point selection for FR2

Subclause	Number of test points	Justification in attachment	Comments
6.2.1 UE maximum output power	x	"38.521-2_TPanalysis_6.2.1_MOP.zip"	RAN5#81
6.4.1 Frequency error	1	"38.521-2_TPanalysis_6.4.1_FreqErr.zip"	RAN5#80
6.4.2.1 Error Vector Magnitude	PUSCH: 168 PUCCH: 24 PRACH: 24	"38.521-2_TPanalysis_6.4.2.1_EVM.zip"	RAN5#3-5G-NR Adhoc
6.4.2.2 Carrier leakage	3	"38.521-2_TPanalysis_6.4.2.2_CarrLeak.zip"	RAN5#3-5G-NR Adhoc
6.4.2.3 In-band emissions	PUSCH: 36 PUCCH: 18	"38.521-1_TPanalysis_6.4.2.3_IE.zip"	RAN5#3-5G-NR Adhoc
6.4.2.4 EVM equalizer spectrum flatness	18	"38.521-2_TPanalysis_6.4.2.4_6.4.2.5_EVMequalizerSpectrumFlatness.zip"	RAN5#3-5G-NR Adhoc
6.4.2.5 EVM spectral flatness for pi/2 BPSK modulation with spectrum shaping	9	"38.521-2_TPanalysis_6.4.2.4_6.4.2.5_EVMequalizerSpectrumFlatness.zip"	RAN5#3-5G-NR Adhoc
6.5.1 Occupied Bandwidth	12	"38.521-2_TPanalysis_6.5.1_OccBW.zip"	RAN5#2-5G-NR Adhoc
6.5.2.1 Spectrum Emission Mask	90	"38.521-2_TPanalysis_6.5.2.1_SEM.zip"	RAN5#2-5G-NR Adhoc RAN5#79 RAN5#80
6.5.2.3 Adjacent Channel Leakage Ratio	TBD	"38.521-2_TPanalysis_6.5.2.3_ACLR.zip"	RAN5#2-5G-NR Adhoc
6.5.3 Spurious emissions	2	"38.521-2_TPanalysis_6.5.3_TxSpurious.zip"	RAN5#80

Table 4.2-2: NR UE receiver test point selection for FR2

Subclause	Number of test points	Justification in attachment	Comments
7.3 Reference sensitivity	9	"38.521-2_TPanalysis_7.3_RefSense.zip"	RAN5#80
7.4 Maximum input level	3	"38.521-2_TPanalysis_7.4_Maximun input level.zip"	RAN5#81
7.5 Adjacent channel selectivity	3	"38.521-2_TPanalysis_7.5 ACS.zip"	RAN5#2-5G-NR Adhoc
7.6.2 In Band Blocking	3	"38.521-2_TPanalysis_7.6.2 InB_Block.zip"	RAN5#2-5G-NR Adhoc

4.3 Test point analysis for NR CA and EN-DC

This clause contains information on test point analysis and test point selection for test cases in [4] clause 6 and 7 with information about transmitting test point selection for NR CA and EN-DC listed in table 4.3-1 and receiver test point selection in table 4.3-2.

Table 4.3-1: NR UE transmitter test point selection for NR CA and EN-DC

Subclause	Number of test points	Justification in attachment	Comments
6.2.1 UE maximum output power	TBD	TBD	TBD
6.2.2 Maximum Power Reduction (MPR)	TBD	TBD	TBD
6.2B.2.3 UE Maximum Output Power reduction for Inter-Band EN-DC within FR1	Same as Table 4.1-1, test case 6.5.2.	Same as Table 4.1-1, test case 6.5.2.	RAN5#3-5G-NR Adhoc
6.2B.2.4 UE Maximum Output Power reduction for Inter-Band EN-DC including FR2	Same as Table 4.2-1, test case 6.2.2	Same as Table 4.2-1, test case 6.2.2	RAN5#81
6.2B.3.1 UE Additional Maximum Output Power reduction for Intra-band contiguous EN-DC	304	"38.521-3_TPanalysis_6.2B.3.1_AMPR_NS_04_v2.zip"	RAN5#81
	8	"38.521-3_TPanalysis_6.2B.3.1_AMPR_NS_35.zip"	RAN5#3-5G-NR Adhoc
6.5B.1.1 Occupied bandwidth for Intra-Band Contiguous EN-DC	X= intraband ENDC channel BWs supported by UE	"38.521-3_TPanalysis_6.5B.1.1_OBW_Intra_B_contig.zip"	RAN5#3-5G-NR adhoc
6.5B.2.1.1 Spectrum emissions mask for intra-band contiguous EN-DC	304	"38.521-3_TPanalysis_6.5B.2.1.1_SEM_Intra_B_contig.zip"	RAN5#3-5G-NR adhoc
6.5B.3.1 Spurious Emissions for intra-band contiguous EN-DC	1	"38.521-3_TPanalysis_6.5B.3.1 TxSpurious_Intra_B_contig.zip"	RAN5#80
6.5B.3.1.2 Spurious emission band UE co-existence for intra-band contiguous EN-DC	108	"38.521-3_TP analysis_6.5B.3.1.2_TX spurious_EN-DC with FR1.zip"	RAN5#3-5G-NR Adhoc

Table 4.3-2: NR UE receiver test point selection for NR CA and EN-DC

Subclause	Number of test points	Justification in attachment	Comments
7.3B.2.1 Reference sensitivity for Intra-band Contiguous EN-DC	45	"38.521-3_TP analysis_7.3B.2.1_RxSense_Intra-band Contiguous EN-DC with FR1.zip"	RAN5#35G-NR Adhoc
7.3B.2.3 Reference sensitivity for Inter-band EN-DC within FR1	45	"38.521-3_TP analysis_7.3B.2.3_RxSense_ Inter-band EN-DC with FR1.zip"	RAN5#35G-NR Adhoc
7.8B.2.3 Wideband Intermodulation for inter-band EN-DC within FR1	Same as Table 4.1-2, test case 7.8.2.	Same as Table 4.1-2, test case 7.8.2.	RAN5#81
7.9B.3 Spurious Emissions for inter-band EN-DC within FR1	Same as Table 4.1-2, test case 7.9.	Same as Table 4.1-2, test case 7.9.	RAN5#81

Annex A: Derivation documents

The documents and spreadsheets used to give the background for the selected test points for each test case are included in the present document as zip files.

The name of the zip shall:

- Include a prefix allowing easier grouping of files in the same area, e.g. “TBD”.
- Include Test Case Number(s), e.g. “TBD”.
- In cases where multiple analysis is needed per test cases, e.g. TBD.

Concatenated example file name: “TBD nnn.zip”.

If there is an update of test points for a test case the old corresponding zip file shall be replaced with a new zip file with a version stepping in the file name. e.g. “nnn_V2.zip”. The aim is to provide a reference to completed test cases, so that test points for similar test cases can be selected on a common basis.

Editor’s note: Rules for naming of zipped background documents will be added to this Annex.

Annex B: Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2017-09	RAN5#76	R5-174704	-	-	-	Draft skeleton TR 38.905	0.0.1
2018-04	RAN5#2-5G-NR Adhoc	R5-181954	-	-	-	Agreed Text Proposal in RAN5#2-5G-NR Adhoc: **R5-181889**, " TP to update TR 38.905 with information on test point analysis " Agreed Test Point Analysis in RAN5#78: **R5-180885**, "Discussion on test point selection for NR Occupied Bandwidth in FR1" **R5-180886**, "Discussion on test point selection for NR SEM in FR1" **R5-180887**, "Discussion on test point selection for NR ACLR in FR1" **R5-181524**, "Discussion on test point selection for Absolute Power Tolerance in FR1" **R5-181525**, "Discussion on test point selection for Aggregate Power Tolerance in FR1" Agreed Test Point Analysis in RAN5#2-5G-NR Adhoc: **R5-182019**, "Discussion of NR FR1 Test Point for TX Spurious Emission test cases " **R5-182024**, "Discussion on test point selection for NR Frequency Error in FR1” **R5-181830**, "Discussion on test point selection for Maximum Output Power in FR1” **R5-181831**, "Discussion on test point selection for Minimum Output Power in FR1**R5-181832**, "Discussion on test point selection for General ON/OFF Time Mask in FR1” **R5-181879**, "Discussion on test point selection for NR In-Band in FR1” **R5-181880**, "Discussion on test point selection for NR ACS in FR1” **R5-182025**, "Discussion on test point selection for NR Frequency Error in FR1” **R5-181905**, "Discussion on test point selection for NR Occupied Bandwidth in FR2” **R5-182030**, "Discussion on test point selection for NR ACLR in FR2” **R5-182042**, "Discussion on test point selection for NR In-Band blocking in FR2” **R5-182044**, "Discussion on test point selection for NR ACS in FR2”	0.1.0
2018-05	RAN5#79	R5-183078	-	-	-	Document title corrected. Agreed Text Proposal in RAN WG5#79: **R5-183963**, "Test Point analysis for FR1 RefSens test case”	0.2.0
2018-08	RAN5#80	R5-185134	-	-	-	**R5-184923**, “Test Point analysis for FR2 RefSense test case” **R5-184961**, “TP for updating TR 38.905 with FR2 Frequency Error test point analysis”	1.0.0

						R5-185307, “TP for updating TR38.905 with FR1 AMPR test point analyses with NS_35” **R5-185309**, “Test Point analysis for FR1 Configured Output Power for SUL” **R5-185311**, “TP for updating TR 38.905 with FR1 Carrier Leakage test point analysis” **R5-185314**, “TP for updating TR 38.905 with FR1 EVM equalizer spectrum flatness test point analysis” **R5-185316**, “TP for updating TR 38.905 with FR1 Frequency Error test point analysis” **R5-185412**, “TP for updating TR 38.905 with EVM test point analysis” **R5-185491**, “Test Point analysis for FR2 TxSpurious test case” **R5-185215**, “TP for updating TR 38.905 with FR2 SEM test point analysis” **R5-185334**, “Discussion of LTE Test point selection for EN-DC with FR1 Tx Spurious emission Test” **R5-185301**, “Discussion on test point selection for NR Out-of-band in FR1” **R5-185423**, “Discussion on Uplink configuration for NR Transmit Intermodulation in FR1” **R5-185216**, “TP for updating TR38.905 with UE AMPR for NS_04 Intra-band contiguous EN-DC” **R5-185319**, “TP for updating TR 38.905 with FR1 In-band Emissions test point analysis”	
2018-09	RAN#81	-	-	-	-	raised to v15.0.0 with editorial changes only	15.0.0
2018-12	RAN#82	R5-186454	0016	-	F	TP analysis for test case 6.5.2.4.2	15.1.0
2018-12	RAN#82	R5-186455	0017	-	F	TP analysis for EN-DC test case 6.2B.2.3	15.1.0
2018-12	RAN#82	R5-186609	0018	-	F	TP_analysis for TX spurious emission UE co-existence for intra-band contiguous EN-DC with FR1	15.1.0
2018-12	RAN#82	R5-186610	0019	-	F	TP analysis for Reference sensitivity for Intra-band Contiguous EN-DC with FR1	15.1.0
2018-12	RAN#82	R5-186611	0020	-	F	TP analysis for Reference sensitivity for Inter-band EN-DC with FR1	15.1.0
2018-12	RAN#82	R5-186674	0021	-	F	Test point analysis for AMPR Intra-band contiguous EN-DC in FR1 for NS_35	15.1.0
2018-12	RAN#82	R5-186710	0022	-	F	TP analysis for test case 6.2B.2.4, UE Maximum Output Power reduction for Inter-Band EN-DC including FR2	15.1.0
2018-12	RAN#82	R5-186791	0028	-	F	TP analysis OBW intraband contiguous EN-DC	15.1.0
2018-12	RAN#82	R5-186792	0029	-	F	TP analysis SEM intraband contiguous EN-DC	15.1.0
2018-12	RAN#82	R5-187035	0031	-	F	Update test points analysis for multiple FR1 test cases	15.1.0
2018-12	RAN#82	R5-187396	0037	-	F	Update of TR 38.905 with SA FR1 A-MPR test point analyses, NS_04	15.1.0
2018-12	RAN#82	R5-188240	0039	1	F	Update of TR 38.905 with EN-DC A-MPR test point analyses, NS_04	15.1.0
2018-12	RAN#82	R5-188227	0041	1	F	Test Point analysis for FR2 Maximum Output Power	15.1.0
2018-12	RAN#82	R5-187489	0042	-	F	TP analysis for FR1 test case 6.3.4.3, relative power tolerance	15.1.0
2018-12	RAN#82	R5-187582	0043	-	F	Discussion on test point selection for EVM in FR2	15.1.0
2018-12	RAN#82	R5-187583	0044	-	F	Discussion on test point selection for Carrier Leakage in FR2	15.1.0
2018-12	RAN#82	R5-187584	0045	-	F	Update of test point selection for EVM equalizer spectrum flatness in FR1	15.1.0
2018-12	RAN#82	R5-187587	0046	-	F	Discussion on test point selection for In-band Emissions in FR2	15.1.0
2018-12	RAN#82	R5-187589	0047	-	F	Discussion on test point selection for EVM equalizer spectrum flatness in FR2	15.1.0
2018-12	RAN#82	R5-187593	0048	-	F	Discussion on test point selection for EVM equalizer spectrum flatness for Pi/2 BPSK in FR1	15.1.0
2018-12	RAN#82	R5-187806	0023	1	F	Test Point analysis for FR1 7.4 Maximum input level	15.1.0
2018-12	RAN#82	R5-187808	0035	1	F	TP analysis for receiver spurious emission tests for FR1 SA	15.1.0
2018-12	RAN#82	R5-187809	0036	1	F	TP analysis for wideband intermodulation tests for FR1 SA	15.1.0
2018-12	RAN#82	R5-187817	0033	1	F	TP analysis for receiver spurious emission tests for FR1 inter-band EN-DC	15.1.0

2018-12	RAN#82	R5-187818	0034	1	F	TP analysis for wideband intermodulation tests for FR1 inter-band EN-DC	15.1.0
2018-12	RAN#82	R5-187836	0025	1	F	Test Point analysis for FR2 7.4 Maximum input level	15.1.0
2018-12	RAN#82	R5-187907	0024	1	F	Test Point analysis for FR1 MPR test case	15.1.0

History

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